

Utilizing Fully Automated Abdominal CT–Based Biomarkers for Opportunistic Screening for Metabolic Syndrome in Adults Without Symptoms

Perry J. Pickhardt, MD¹, Peter M. Graffy, MPH¹, Ryan Zea, MS¹, Scott J. Lee, MD¹, Jiamin Liu, PhD², Veit Sandfort, MD², Ronald M. Summers, MD, PhD²

Gastrointestinal Imaging • Original Research

Keywords

artificial intelligence, CT, deep learning, machine learning, metabolic syndrome

Submitted: Feb 29, 2020

Revision requested: Mar 27, 2020

Revision received: Mar 29, 2020

Accepted: Apr 3, 2020

First published online: Jun 30, 2020

The opinions and assertions contained herein are the private views of the authors and are not to be construed as official or as representing the views of the National Institutes of Health.

P. J. Pickhardt is an advisor to Bracco and Zebra and is a shareholder in SHINE, Elucet Medical, and Cellerar Biosciences. R. M. Summers receives royalties from iCAD, PingAn, Philips and ScanMed and research support from PingAn and NVIDIA. The remaining authors declare that they have no disclosures relevant to the subject matter of this article.

Supported in part by the Intramural Research Program of the National Institutes of Health (NIH) Clinical Center. The high-performance computing capabilities of the NIH Biowulf cluster were used.

OBJECTIVE. Metabolic syndrome describes a constellation of reversible cardiometabolic abnormalities associated with cardiovascular risk and diabetes. The present study investigates the use of fully automated abdominal CT–based biometric measures for opportunistic identification of metabolic syndrome in adults without symptoms.

MATERIALS AND METHODS. International Diabetes Federation criteria were applied to a cohort of 9223 adults without symptoms who underwent unenhanced abdominal CT. After patients with insufficient clinical data for diagnosis were excluded, the final cohort consisted of 7785 adults (mean age, 57.0 years; 4361 women and 3424 men). Previously validated and fully automated CT-based algorithms for quantifying muscle, visceral and subcutaneous fat, liver fat, and abdominal aortic calcification were applied to this final cohort.

RESULTS. A total of 738 subjects (9.5% of all subjects; mean age, 56.7 years; 372 women and 366 men) met the clinical criteria for metabolic syndrome. Subsequent major cardiovascular events occurred more frequently in the cohort with metabolic syndrome ($p < 0.001$). Significant differences were observed between the two groups for all CT-based biomarkers ($p < 0.001$). Univariate L1-level total abdominal fat (area under the ROC curve [AUROC] = 0.909; odds ratio [OR] = 27.2), L3-level skeletal muscle index (AUROC = 0.776; OR = 5.8), and volumetric liver attenuation (AUROC = 0.738; OR = 5.1) performed well when compared with abdominal aortic calcification scoring (AUROC = 0.578; OR = 1.6). An L1-level total abdominal fat threshold of 460.6 cm² was 80.1% sensitive and 85.4% specific for metabolic syndrome. For women, the AUROC was 0.930 when fat and muscle measures were combined.

CONCLUSION. Fully automated quantitative tissue measures of fat, muscle, and liver derived from abdominal CT scans can help identify individuals who are at risk for metabolic syndrome. These visceral measures can be opportunistically applied to CT scans obtained for other clinical indications, and they may ultimately provide a more direct and useful definition of metabolic syndrome.

Metabolic syndrome is characterized by a constellation of potentially reversible cardiometabolic risk factors in the setting of central obesity [1–3]. Measurable clinical components include elevated triglyceride levels, increased blood pressure, an elevated fasting plasma glucose level, and an increased waist circumference (or body mass index [BMI; weight in kilograms divided by the square of height in meters]) as well as a reduced high-density lipoprotein cholesterol level [3]. Metabolic syndrome confers a substantial increased risk for cardiovascular (CV) disease and type 2 diabetes mellitus. The primary goal of identifying individuals with this syndrome is to initiate clinical interventions, especially those centered on weight management, to prevent adverse CV events and diabetes in the future. This highly prevalent condition is expected to become even more prevalent as the obesity epidemic continues to grow [2].

Although both intraabdominal (visceral) fat and ectopic fat (hepatic steatosis) are now widely recognized as being highly relevant to CV risk and metabolic syndrome [1, 4–8], these critical measures are not defining criteria, primarily because they are not readily obtainable through standard clinical (i.e., nonimaging) means. Instead, increased intraabdominal fat is only indirectly assessed via an elevated BMI or enlarged waist circumference, both of which have shortcomings as surrogates for central obesity [1]. Abdominal

This article is available for credit.

doi.org/10.2214/AJR.20.23049

AJR 2021; 216:85–93

ISSN-L 0361–803X/21/2161–85

© American Roentgen Ray Society

¹Department of Radiology, University of Wisconsin School of Medicine & Public Health, E3/311 Clinical Science Center, 600 Highland Ave, Madison, WI 53792-3252. Address correspondence to P. J. Pickhardt (ppickhardt2@uwhealth.org).

²Imaging Biomarkers and Computer-Aided Diagnosis Laboratory, Radiology and Imaging Sciences, National Institutes of Health Clinical Center, Bethesda, MD.

CT is widely performed for a variety of clinical conditions, providing a potential means for opportunistic detection of cardiometabolic risk factors [9]. Furthermore, through artificial intelligence means, quantification of relevant body tissues can be obtained in a fully automated manner, providing rapid and objective measurements [10–14]. The purpose of this study was to evaluate the use of fully automated abdominal CT-based measures of body composition for opportunistic identification of metabolic syndrome in adults without symptoms.

Materials and Methods

Patient Cohort and CT Protocol

This HIPAA-compliant investigation was approved by the institutional review board at the University of Wisconsin School of Medicine & Public Health and the Office of Human Research Protection at the National Institutes of Health Clinical Center. The requirement for signed informed consent was waived for this retrospective assessment.

We applied International Diabetes Federation (IDF) criteria for diagnosing metabolic syndrome to an outpatient cohort of 9223 generally healthy consecutive adults without symptoms who underwent low-dose unenhanced abdominal CT for routine colorectal cancer screening between 2004 and 2016 at a single medical center [3]. These IDF criteria, which are perhaps the most widely accepted criteria for diagnosing metabolic syndrome, require the presence of either an increased waist circumference (with sex-specific thresholds) or a BMI of 30 or higher plus either at least two of the three following findings—elevated triglyceride levels, increased blood pressure, or an elevated fasting plasma glucose level—or a reduced high-density lipoprotein cholesterol level. For the present study, we used BMI as the clinical criterion for obesity. However, it is now recognized that both waist circumference and BMI are indirect measures and are somewhat inaccurate for assessing central obesity [1]. Although the individual defining thresholds used are below treatment levels, they double the risk for coronary heart disease when combined [2]. We also performed a systematic search of the electronic health record for major CV events (e.g., myocardial infarction, cerebrovascular accident, or congestive heart failure) or death occurring after the time that CT was performed. In addition, we searched for a formal diagnosis of type 2 diabetes mellitus made at any point in time. We excluded individuals who had either inadequate clinical data for ascertaining metabolic syndrome status or an inadequate clinical follow-up (duration of < 1 year) in the absence of an adverse event.

For this investigation, low-dose unenhanced MDCT was performed with the patient in the supine position, and all scans were obtained as part of a colonography screening examinations conducted using a scanner from a single vendor. Parameters included a peak kilovoltage of 120 kVp and tube current modulation to achieve a noise index of 50, typically resulting in an effective dose of 2–3 mSv. The specific additional CT colonography-related techniques for bowel preparation and colonic distention are beyond the scope of this investigation but have been previously described elsewhere [15, 16].

Automated CT Biomarkers

The deep learning and image processing algorithms used in the present study were all previously developed, trained, and tested, including internal and external validation tests, at the

University of Wisconsin School of Medicine & Public Health [17–26]. We have published separate technical studies describing the success rates, normative values, and changes seen over time with the use of these automated CT-based tools [10–13]. No additional training, validation, or machine learning was required for this study, which obviated the need to repeat separate training and testing cohorts.

Although the technical details regarding the artificial intelligence methods used for automated CT-based anatomic tissue segmentation and quantification have been enumerated elsewhere, a brief description follows. For abdominal fat quantification, feature-based image processing algorithms first segment and label the spinal levels from T12 to L5; this is followed by isolation of the visceral and subcutaneous fat compartments at each vertebral level. For the abdominal wall musculature, liver, and calcified atherosclerotic aortic plaque quantification, deep learning algorithms were used, including a modified convolutional neural network (3D U-Net) for segmentation of liver and muscle and the mask region-based convolutional neural network (R-CNN) algorithm for segmentation of aortic calcium.

On the basis of our prior research, we focused on fat quantification at the L1 level [10], muscle quantification at the L3 level [12], abdominal aortic calcification quantification at the L1–L4 levels [11], and liver fat quantification utilizing the entire available liver volume [13]. These body composition measures can be reported in a number of different ways. For the present study, we quantified fat as visceral, subcutaneous, and total adipose tissue (VAT, SAT, and TAT, respectively) according to the cross-sectional area at the L1 level (TAT = VAT + SAT). Muscle measures included mean attenuation (expressed as Hounsfield units) and the cross-sectional area at the L3 level, in addition to the skeletal muscle index, which was defined as the L3 cross-sectional area (expressed as square centimeters) divided by patient height (expressed in meters). Abdominal aortic calcification (inclusive of levels L1–L4) was reported in terms of the Agatston score, which is the standard weighted score based on attenuation values originally developed for coronary calcium CT. Finally, liver fat was quantified on the basis of mean volumetric attenuation (expressed as Hounsfield units), which has a linear correlation with the MRI-based proton density fat fraction reference standard [17, 18]. Figure 1 shows visual correlates of the quantitative output for these automated CT tools.

Statistical Analysis

Summary statistics were compiled and compared for individuals with and without metabolic syndrome. Relevant *p* values were derived using two-sided *t* tests for normally distributed variables and using the Wilcoxon rank sum test when the normality assumption did not hold; *p* < 0.01 was used to determine statistical significance. Odds ratios (ORs) with 95% CIs were computed for each CT biomarker, with the highest-risk quartile compared with the other three quartiles. ROC curve analysis was used to assess the association between CT-based body composition measures and metabolic syndrome. The area under the ROC curve (AUROC) values with 95% CIs were calculated for univariate and multivariate measures for the entire cohort and according to sex. Diagnostic performance (sensitivity and specificity) was calculated at thresholds that resulted in either 80% sensitivity or 80% specificity. Because BMI is a central defining criterion for metabolic syndrome, predictive results according to BMI cannot be reported or compared.

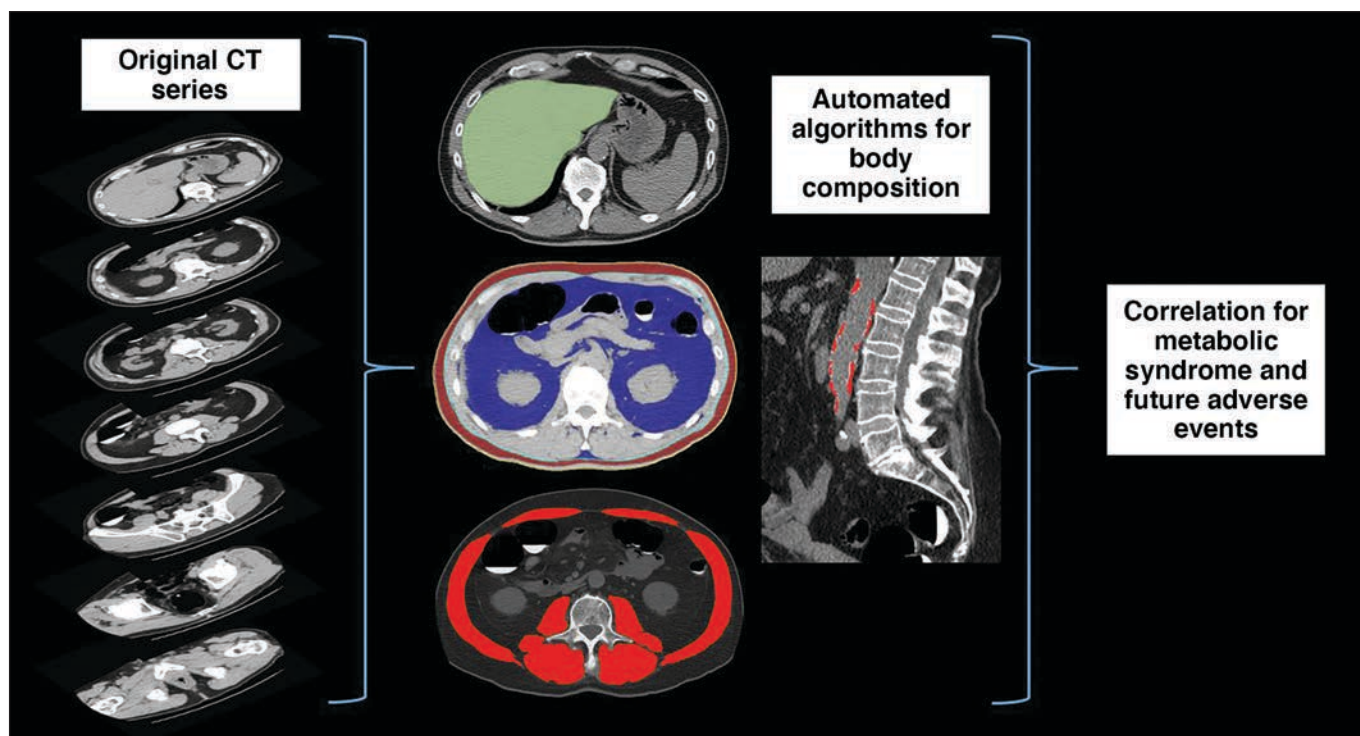


Fig. 1—Schematic shows fully automated CT biomarker tools for assessing liver, fat, muscle, and abdominal aortic calcium on basis of original abdominal CT scan data. CT images depicting automated liver, fat, and muscle tools are from 66-year-old man with metabolic syndrome. Sagittal image depicting aortic calcium tool is from different patient. CT biomarker results were correlated with clinical diagnosis of metabolic syndrome and future major cardiovascular events and death.

Results

After exclusions, the final cohort consisted of 7785 adults without symptoms (mean age, 57.0 years; 4361 women and 3424 men). Of these, 738 subjects (9.5%; mean age, 56.7 years; 372 women and 366 men) met IDF clinical criteria for metabolic syndrome, and 7047 (90.5%; mean age, 57.1 years; 3989 women and 3058 men) did not (Table 1). The median duration of clinical follow-up after CT was 8.6 years (interquartile range [IQR], 5.1–11.4 years; maximum, 14.0 years). Subsequent major CV events occurred more frequently ($p < 0.001$) in the cohort with metabolic syndrome (20.9%; 154/738) than in those without metabolic syndrome (15.6%; 1096/7047). However, no significant difference in overall survival

was observed during the clinical follow-up interval (5.3% of those with metabolic syndrome died versus 5.7% of those without metabolic syndrome). A total of 2156 individuals (27.7%; 1101 women and 1055 men) were given a specific diagnosis of diabetes.

For all reported CT-based biomarkers, significant differences ($p < 0.001$) were observed between individuals with and without metabolic syndrome. Mean values (with SDs) and IQRs are presented in Table 2. The mean VAT for subjects with metabolic syndrome was more than double that for those without (303.3 vs 140.1 cm²), whereas differences in SAT (and TAT) were less pronounced. For both TAT and VAT, no overlap in IQRs existed between individuals with and without metabolic syndrome. On

TABLE 1: Summary of Demographic and Clinical Characteristics According to Presence of Metabolic Syndrome

Characteristic	Total Cohort (<i>n</i> = 7785)	Subjects With Metabolic Syndrome (<i>n</i> = 738)	Subjects Without Metabolic Syndrome (<i>n</i> = 7047)
Patient age (y)			
Mean ± SD	57.0 ± 7.9	56.7 ± 6.9	57.1 ± 8.0
IQR	51.0–61.0	51.0–61.0	51.0–61.0
Female sex	4361 (56.0)	372 (50.4)	3989 (56.6)
Subsequent adverse outcome			
Major cardiovascular event	1250 (16.1)	154 (20.9)	1096 (15.6)
Death	444 (5.7)	39 (5.3)	405 (5.7)

Note—Except where otherwise indicated, data are number (%) of subjects. IQR = interquartile range.

TABLE 2: Summary of Automated CT Biomarker Data According to Presence of Metabolic Syndrome

CT Biomarker	Total Cohort (n = 7785)	Subjects With Metabolic Syndrome (n = 738)	Subjects Without Metabolic Syndrome (n = 7047)
Total adipose tissue			
Mean ± SD	343.7 ± 172.5	596.0 ± 164.5	317.1 ± 150.4
IQR	221.8–434.5	478.2–685.2	209.9–400.2
Visceral adipose tissue			
Mean ± SD	155.6 ± 110.4	303.3 ± 146.3	140.1 ± 93.2
IQR	75.7–205.3	200.1–357.3	71.2–183.3
Subcutaneous adipose tissue			
Mean ± SD	191.2 ± 96.9	302.1 ± 107.4	179.5 ± 87.9
IQR	123.4–242.0	222.4–369.0	117.7–227.3
Skeletal muscle index			
Mean ± SD	51.4 ± 12.0	62.3 ± 12.8	50.3 ± 11.3
IQR	43.5–59.0	54.0–70.4	43.0–57.4
Liver attenuation (HU)			
Mean ± SD	56.3 ± 9.6	47.3 ± 13.9	57.3 ± 8.5
IQR	53.0–62.0	39.0–57.0	54.0–62.0
Abdominal aortic Agatston score			
Mean ± SD	668.7 ± 1705.7	950.1 ± 2064.1	639.0 ± 1661.1
IQR	0.0–455.5	0.0–823.2	0.0–426.0

Note—All comparisons for automated CT biomarkers were statistically significant ($p < 0.001$) between individuals with metabolic syndrome versus those without metabolic syndrome. Note the lack of overlap in the interquartile range (IQR) for total adipose tissue and subcutaneous adipose tissue.

average, those with metabolic syndrome had moderate hepatic steatosis (47.3 HU), whereas those without metabolic syndrome had normal mean liver attenuation (57.3 HU).

Table 3 shows the diagnostic performance for predicting IDF-defined metabolic syndrome according to ROC analysis, and

Figure 2 shows selected ROC curves. ORs are also presented in Table 3. For L1-level TAT, the univariate AUROC was 0.909 (0.922 for women and 0.891 for men) with an associated OR of 27.2 (95% CI, 22.0–34.1). Likewise, for the L3-level skeletal muscle index, the AUROC was 0.776 with an OR of 5.8 (IQR, 5.0–6.8); and for

TABLE 3: ROC Analysis and Odds Ratios of CT Biomarkers for Predicting Metabolic Syndrome

CT Biomarker	AUROC			Odds Ratio
	For Men (n = 3424)	For Women (n = 4361)	For All Subjects (n = 7785)	
Univariate performance				
TAT	0.891	0.922	0.909	27.2
VAT	0.857	0.902	0.862	10.6
SAT	0.808	0.874	0.823	8.1
SMI	0.786	0.817	0.776	5.8
Muscle area	0.786	0.812	0.729	3.2
Muscle attenuation	0.619	0.697	0.635	2.5
Liver attenuation	0.730	0.743	0.738	5.1
Abdominal aortic Agatston score	0.548	0.593	0.578	1.6
Multivariate performance				
TAT and SMI	0.904	0.930	0.916	—
TAT and liver attenuation	0.894	0.924	0.911	—
TAT, SMI, and liver attenuation	0.908	0.933	0.919	—

Note—Dash (—) denotes not applicable. AUROC = area under the ROC curve, TAT = total adipose tissue, VAT = visceral adipose tissue, SAT = subcutaneous adipose tissue, SMI = skeletal muscle index, AoCa = aortic calcification.

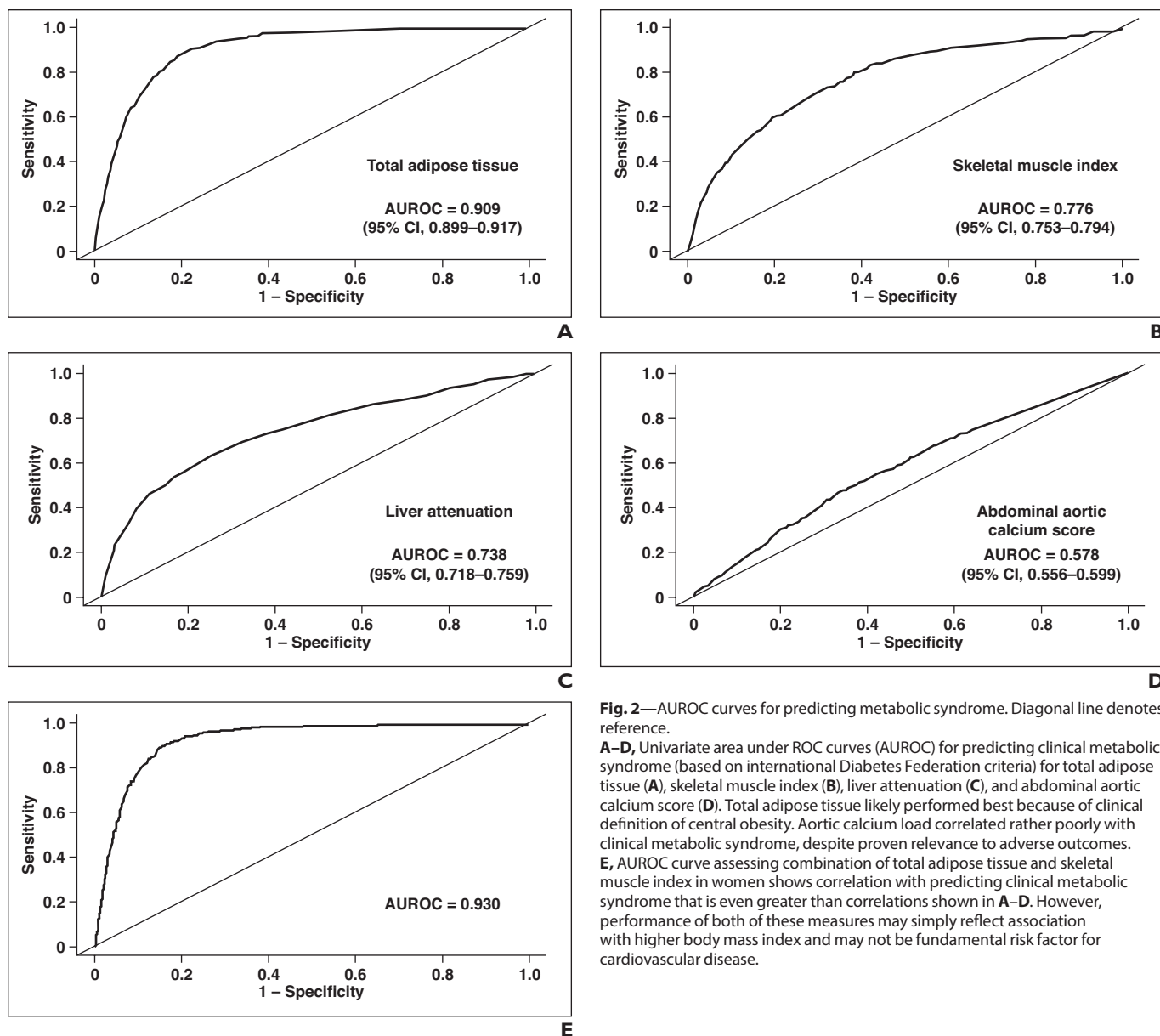


Fig. 2—AUROC curves for predicting metabolic syndrome. Diagonal line denotes reference.

A–D, Univariate area under ROC curves (AUROC) for predicting clinical metabolic syndrome (based on international Diabetes Federation criteria) for total adipose tissue (**A**), skeletal muscle index (**B**), liver attenuation (**C**), and abdominal aortic calcium score (**D**). Total adipose tissue likely performed best because of clinical definition of central obesity. Aortic calcium load correlated rather poorly with clinical metabolic syndrome, despite proven relevance to adverse outcomes.

E, AUROC curve assessing combination of total adipose tissue and skeletal muscle index in women shows correlation with predicting clinical metabolic syndrome that is even greater than correlations shown in **A–D**. However, performance of both of these measures may simply reflect association with higher body mass index and may not be fundamental risk factor for cardiovascular disease.

volumetric liver attenuation, the AUROC was 0.738 with an OR of 5.1 (IQR, 4.4–6.0). In contrast, the L1–L4 abdominal aortic calcification score underperformed, with an AUROC of 0.578 and an OR of 1.6 (IQR, 1.3–1.8). For the multivariate combination of fat and muscle measures (i.e., TAT plus skeletal muscle index) in women, the AUROC was 0.930. For correlation with a diagnosis of diabetes, the CT-based L1-level visceral fat measure (VAT) showed the highest overall performance (AUROC = 0.702), topping all other CT measures as well as BMI.

Table 4 reports the diagnostic performance of specific thresholds for the CT-based measures. For example, with use of a TAT threshold of 460.6 cm², the sensitivity and specificity for detection of metabolic syndrome were 80.1% and 85.4%, respectively. Of note, this performance is based on the use of clinical diagnosis for central obesity and incorporates neither visceral nor hepatic fat in the reference standard. Figure 3 shows an individual case example that illustrates this definitional disconnect because this

patient did not qualify as having metabolic syndrome according to the clinical IDF criteria but nonetheless had abnormal visceral and hepatic fat measures as well as a poor clinical outcome.

Discussion

Cardiometabolic risk factors affecting overweight individuals, including dyslipidemia, hypertension, and a high fasting plasma glucose level, confer a greatly increased risk of type 2 diabetes and premature CV disease and are known as metabolic syndrome. With the ongoing global obesity epidemic, the prevalence of metabolic syndrome is expected to rise. It is estimated that metabolic syndrome now affects up to 30–40% of people by the time they are 65 or 70 years old [2, 19]. The prevalence was notably lower in our younger and generally healthy screening cohort, perhaps in part because BMI was used as the main criterion for identifying central obesity. However, even a prevalence of 10% is actionable. It is critical to identify at-risk individuals before they have symptoms

TABLE 4: Diagnostic Performance of Fully Automated CT Biomarkers for Predicting Metabolic Syndrome

CT Biomarker and Threshold ^a	Sensitivity (%)	Specificity (%)
Total adipose tissue		
460.6 cm ²	80.1	85.4
426.4 cm ²	88.3	80.0
Visceral adipose tissue		
186.9 cm ²	80.1	74.7
205.4 cm ²	72.7	80.0
Skeletal muscle index		
52.8	80.3	61.8
59.4	60.3	80.1
Liver attenuation		
59.5 HU	82.5	44.3
52.5 HU	56.1	80.5

^aThresholds were selected to provide approximately 80% sensitivity or specificity.

of metabolic syndrome develop, especially given the fact that the course of this condition is potentially reversible through standard clinical interventions and lifestyle modifications.

In the clinic, targeting individuals with a protuberant, apple-shaped abdomen seems like an obvious starting point, but this approach may lack both sensitivity and specificity in terms of the true underlying pathologic condition. Clinical substitutes for BMI and abdominal circumference have been studied [20], but they fall short of imaging-based visceral measures. Disproportionate increases in visceral and hepatic fat more directly reflect an even higher risk profile [1, 4], but they require cross-sectional imaging for identification and quantification. However, given the high rate of utilization of abdominal CT for a wide variety of clinical indications in the United States [21] and many other developed nations, opportunistic detection of these critical risk factors can be achieved. In a prior, smaller CT-based study, we assessed manual and semiautomated methods for quantifying liver and visceral fat [4]. In the present study, we report a fully automated approach for quantifying even more body composition parameters that are relevant to metabolic syndrome in a much larger cohort.

Because the clinical criteria acting as surrogates for central obesity in metabolic syndrome are relatively crude, we cannot measure the true impact of CT-based visceral and hepatic fat quantification in our cohort. It is entirely possible that this flawed clinical reference standard for central obesity is obscuring the relative importance of direct measurements of visceral and hepatic fat, as well as other CT-based measures of body composition. To wit, we have previously shown that CT-based quantification of visceral fat, liver fat, muscle, and abdominal aortic calcification are all effective predictors of future major CV events and death [9], ignoring the clinical definition of metabolic syndrome. In comparison, subjects with metabolic syndrome in our study did not show decreased survival, which was surprising and likely indicates weakness in the clinically based definition of central obesity. Other studies have also shown improved prediction of rele-

vant outcomes when using CT-based visceral measures [1, 4–8]. Furthermore, the predictive value of CT-based abdominal aortic calcification scoring alone surpasses that of the multivariate clinically based Framingham Risk Score [9, 22], which includes many of the same clinical features found in criteria for metabolic syndrome. As such, it was also somewhat surprising that abdominal aortic calcification scoring did not correlate well with metabolic syndrome, especially given its association with diabetes. Regardless, the overall ability of objective CT-based measures to identify those at risk for major CV events (and death) trumps that of an arbitrary clinical definition of the metabolic syndrome. These collective findings suggest the need for an improved definition of the metabolic syndrome that incorporates these imaging biomarkers that reflect important visceral changes in body composition. We next aim to determine optimal thresholds for an improved CT-based definition of metabolic syndrome and apply this to a novel patient cohort.

The fact that TAT measures (i.e., VAT plus SAT measures) outperformed VAT alone also likely reflects the aforementioned shortcomings related to the BMI and waist circumference criteria that are unavoidable components of the clinical reference standard. According to some studies, metabolic syndrome may actually be promoted by a lack of SAT [2]. Another result that appears nonintuitive is that increased skeletal muscle mass correlated with metabolic syndrome in the present study. This finding runs counter to some observations that low skeletal muscle mass may promote metabolic syndrome [2]. However, other researchers have argued that skeletal muscle resistance to insulin is more likely to contribute to the development of metabolic syndrome [23]. Nonetheless, the simplest explanation might be that individuals with more muscle mass have a larger BMI and therefore are more likely to qualify for metabolic syndrome, perhaps artificially so. Whether those with sarcopenic obesity are an important subset of at-risk individuals is uncertain, but our study found no support for this. In contrast, lower muscle attenuation correlated with metabolic syndrome. Others have shown that lower visceral fat attenuation also correlates with inflammatory markers and cardiometabolic risk [8].

The present study provides further evidence for the potential value of harnessing the rich body composition data that are embedded within all abdominal CT scans and typically are unused in routine practice. The current investigation focused on the opportunistic detection of metabolic syndrome using automated quantification of fat, muscle, liver, and abdominal aortic calcium. We and others have used these visceral measures to identify individuals at greater risk for future major CV events and overall survival [9, 24]. By adding bone assessment, we have also assessed the ability to predict future fragility fractures [25]. In addition to automated measures, manual approaches to CT-based opportunistic screening can also be applied, often on the fly [26–33]. These assessments that are incidental to the clinical indication for scanning have the potential to add substantial value to CT performed for a wide variety of other reasons. Furthermore, when these opportunistic screening efforts are coupled with an established indication such as CT colonography for colorectal cancer prevention, this further supports the concept of stand-alone population-based CT screening of adults without symptoms.

We acknowledge the limitations of the present study. As already discussed, the most critical limitation of this study is related to the indirect (and inaccurate) clinical definition of central

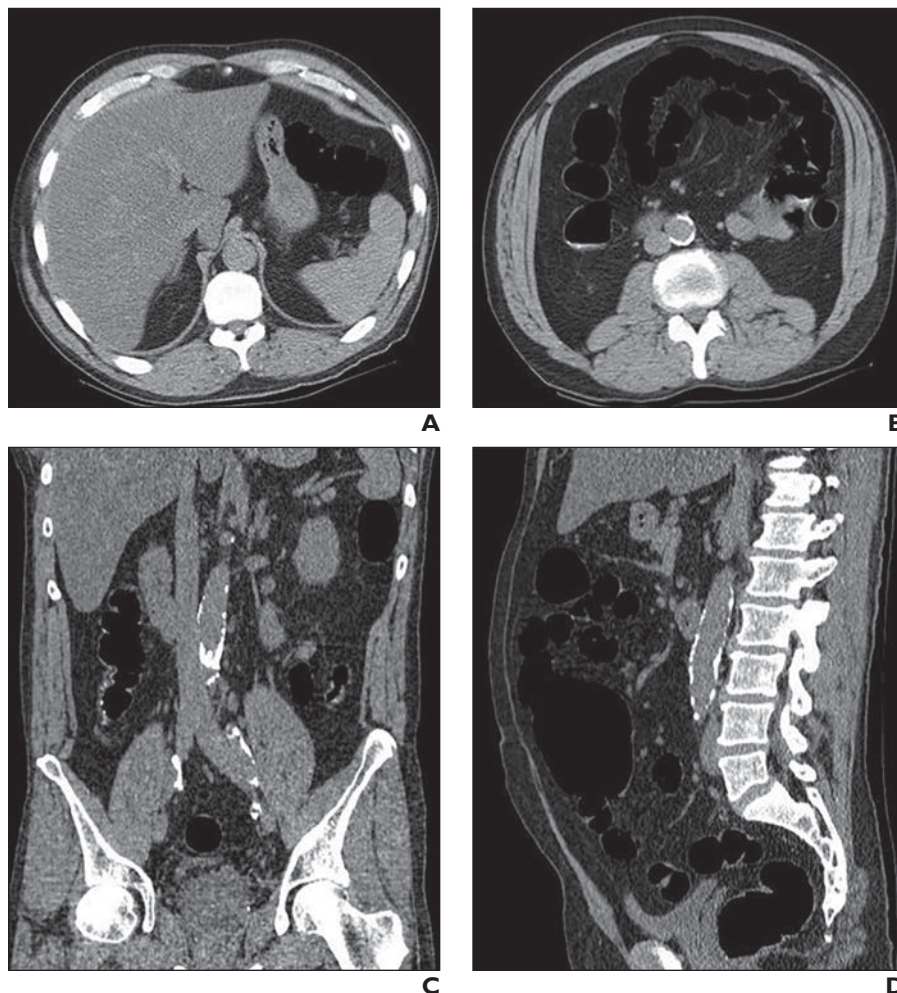


Fig. 3—52-year-old man with metabolic syndrome without symptoms. Patient did not meet criteria for metabolic syndrome (body mass index [weight in kilograms divided by the square of height in meters], 27.3; triglyceride level, 78 mg/dL; high-density lipoprotein cholesterol level, 51 mg/dL) but had positive findings for hypertension. His CT-based liver fat, visceral fat, and abdominal aortic calcification scores were all above 95th percentile for entire cohort, suggesting metabolic derangement. He had myocardial infarction develop 3 years after undergoing CT, had cerebrovascular accident 5 years after myocardial infarction, and died 4 years after cerebrovascular accident. This information argues for inclusion of visceral and hepatic fat in definition of metabolic syndrome.

A–D, Unenhanced CT scans show hepatic steatosis (**A**), prominent visceral fat (**B**), and extensive aortic calcification (**B–D**).

obesity used to establish a diagnosis of metabolic syndrome. For this reason, we also could not compare CT-based performance with BMI because the latter was used as a defining criterion. This flawed reference standard very likely underestimated the performance of the CT-based visceral measures. A new definition incorporating these visceral measures might further improve risk stratification. Another limitation was the necessary exclusion of individuals with insufficient clinical data for determining the presence or absence of metabolic syndrome. Because this was a single-center study, possible selection bias exists; additional investigation of other more diverse cohorts is warranted. We used CT scans obtained with a low-dose colonography technique, but we anticipate that the results would generally hold for scans obtained using the routine-dose standard technique. Finally, although results of artificial intelligence–based study generally should be viewed with caution because of the black box nature of some algorithms, this concern was largely averted because we used established, static tools without any additional learning.

In conclusion, fully automated quantitative tissue measures of fat, muscle, and liver derived from abdominal CT scans can identify individuals at risk for metabolic syndrome. These measures can be opportunistically applied to CT scans obtained for other clinical indications, possibly leading to unsuspected detection and potential intervention of this important yet reversible condition. Perhaps even more importantly, utilizing these CT-based visceral measures could potentially lead to a more relevant and fundamental definition of the metabolic syndrome.

References

1. Neeland IJ, Ross R, Després JP, et al.; International Atherosclerosis Society; International Chair on Cardiometabolic Risk Working Group on Visceral Obesity. Visceral and ectopic fat, atherosclerosis, and cardiometabolic disease: a position statement. *Lancet Diabetes Endocrinol* 2019; 7:715–725
2. Han TS, Lean MEJ. A clinical perspective of obesity, metabolic syndrome and cardiovascular disease. *JRSM Cardiovasc Dis* 2016; 5:2048004016633371
3. Alberti KG, Zimmet P, Shaw J. Metabolic syndrome: a new world-wide definition—a consensus statement from the International Diabetes Federation. *Diabet Med* 2006; 23:469–480
4. Pickhardt PJ, Jee Y, O'Connor SD, del Rio AM. Visceral adiposity and hepatic steatosis at abdominal CT: association with the metabolic syndrome. *AJR* 2012; 198:1100–1107
5. Lonardo A, Ballestri S, Marchesini G, Angulo P, Loria P. Nonalcoholic fatty liver disease: a precursor of the metabolic syndrome. *Dig Liver Dis* 2015; 47:181–190
6. Graffy PM, Pickhardt PJ. Quantification of hepatic and visceral fat by CT and MR imaging: relevance to the obesity epidemic, metabolic syndrome and NAFLD. *Br J Radiol* 2016; 89:20151024
7. Goldenberg L, Saliba W, Hayeq H, Hasadia R, Zeina AR. The impact of abdominal fat on abdominal aorta calcification measured on non-enhanced CT. *Medicine (Baltimore)* 2018; 97:e13233
8. Shah RV, Allison MA, Lima JA, et al. Abdominal fat radiodensity, quantity and cardiometabolic risk: The Multi-Ethnic Study of Atherosclerosis. *Nutr Metab Cardiovasc Dis* 2016; 26:114–122
9. Pickhardt PJ, Graffy PM, Zea R, et al. Automated CT biomarkers for opportunistic prediction of future cardiovascular events and mortality in an asymptomatic screening population. *Lancet Digit Health* 2020; 2:e192–e200

10. Lee SJ, Liu J, Yao J, Kanarek A, Summers RM, Pickhardt PJ. Fully automated segmentation and quantification of visceral and subcutaneous fat at abdominal CT: application to a longitudinal adult screening cohort. *Br J Radiol* 2018; 91:20170968
11. Graffy PM, Liu J, O'Connor S, Summers RM, Pickhardt PJ. Automated segmentation and quantification of aortic calcification at abdominal CT: application of a deep learning-based algorithm to a longitudinal screening cohort. *Abdom Radiol (NY)* 2019; 44:2921–2928
12. Graffy PM, Liu J, Pickhardt PJ, Burns JE, Yao J, Summers RM. Deep learning-based muscle segmentation and quantification at abdominal CT: application to a longitudinal adult screening cohort for sarcopenia assessment. *Br J Radiol* 2019; 92:20190327
13. Graffy PM, Sandfort V, Summers RM, Pickhardt PJ. Automated liver fat quantification at nonenhanced abdominal CT for population-based steatosis assessment. *Radiology* 2019; 293:334–342
14. Pickhardt PJ, Lee SJ, Liu J, et al. Population-based opportunistic osteoporosis screening: validation of a fully automated CT tool for assessing longitudinal BMD changes. *Br J Radiol* 2019; 92:20180726
15. Pickhardt PJ, Choi JR, Hwang I, et al. Computed tomographic virtual colonoscopy to screen for colorectal neoplasia in asymptomatic adults. *N Engl J Med* 2003; 349:2191–2200
16. Kim DH, Pickhardt PJ, Taylor AJ, et al. CT colonography versus colonoscopy for the detection of advanced neoplasia. *N Engl J Med* 2007; 357:1403–1412
17. Guo Z, Blake GM, Li K, et al. Liver fat content measurement with quantitative CT validated against MRI proton density fat fraction: a prospective study of 400 healthy volunteers. *Radiology* 2020; 294:89–97
18. Pickhardt PJ, Graffy PM, Reeder SB, Hernando D, Li K. Quantification of liver fat content with unenhanced MDCT: phantom and clinical correlation with MRI proton density fat fraction. *AJR* 2018; 211:[web]W151–W157
19. Ford ES, Giles WH, Mokdad AH. Increasing prevalence of the metabolic syndrome among U.S. Adults. *Diabetes Care* 2004; 27:2444–2449
20. Kwon S, Han AL. The correlation between the ratio of visceral fat area to subcutaneous fat area on computed tomography and lipid accumulation product as indexes of cardiovascular risk. *J Obes Metab Syndr* 2019; 28:186–193
21. [No authors listed]. 2018 CT market outlook report. Des Plaines, IL: IMV Medical Information Division, 2018
22. O'Connor SD, Gsraffy PM, Zea R, Pickhardt PJ. Does nonenhanced CT-based quantification of abdominal aortic calcification outperform the Framingham risk score in predicting cardiovascular events in asymptomatic adults? *Radiology* 2019; 290:108–115
23. Stump CS, Henriksen EJ, Wei Y, Sowers JR. The metabolic syndrome: role of skeletal muscle metabolism. *Ann Med* 2006; 38:389–402
24. Shah RV, Yeri AS, Murthy VL, et al. Association of multiorgan computed tomographic phenomap with adverse cardiovascular health outcomes: the Framingham Heart Study. *JAMA Cardiol* 2017; 2:1236–1246
25. Pickhardt PJ, Graffy PM, Zea R, et al. Automated abdominal CT imaging biomarkers for opportunistic prediction of future major osteoporotic fractures in asymptomatic adults. *Radiology* 2020; 297:64–72
26. Jang S, Graffy PM, Ziemlewicz TJ, Lee SJ, Summers RM, Pickhardt PJ. Opportunistic osteoporosis screening at routine abdominal and thoracic CT: normative L1 trabecular attenuation values in more than 20 000 adults. *Radiology* 2019; 291:360–367
27. Lee SJ, Pickhardt PJ. Opportunistic screening for osteoporosis using body CT scans obtained for other indications: the UW experience. *Clin Rev Bone Miner Metab* 2017; 15:128–137
28. Ziemlewicz TJ, Maciejewski A, Binkley N, Brett AD, Brown JK, Pickhardt PJ. Opportunistic quantitative CT bone mineral density measurement at the proximal femur using routine contrast-enhanced scans: direct comparison with DXA in 355 adults. *J Bone Miner Res* 2016; 31:1835–1840
29. Ziemlewicz TJ, Binkley N, Pickhardt PJ. Opportunistic osteoporosis screening: addition of quantitative CT bone mineral density evaluation to CT colonography. *J Am Coll Radiol* 2015; 12:1036–1041
30. Pickhardt PJ, Pooler BD, Lauder T, del Rio AM, Bruce RJ, Binkley N. Opportunistic screening for osteoporosis using abdominal computed tomography scans obtained for other indications. *Ann Intern Med* 2013; 158:588–595
31. Hahn L, Reeder SB, Muñoz del Rio A, Pickhardt PJ. Longitudinal changes in liver fat content in asymptomatic adults: hepatic attenuation on unenhanced CT as an imaging biomarker for steatosis. *AJR* 2015; 205:1167–1172
32. Boyce CJ, Pickhardt PJ, Kim DH, et al. Hepatic steatosis (fatty liver disease) in asymptomatic adults identified by unenhanced low-dose CT. *AJR* 2010; 194:623–628
33. Lawrence EM, Pooler BD, Pickhardt PJ. Opportunistic screening for hereditary hemochromatosis with unenhanced CT: determination of an optimal liver attenuation threshold. *AJR* 2018; 211:1206–1211

FOR YOUR INFORMATION

ARRS is accredited by the Accreditation Council for Continuing Medical Education (ACCME) to provide continuing medical education activities for physicians.

The ARRS designates this journal-based CME activity for a maximum of 1.00 *AMA PRA Category 1 Credits™* and 1.00 *American Board of Radiology®*, MOC Part II, Self-Assessment CME (SA-CME). Physicians should claim only the credit commensurate with the extent of their participation in the activity.

To access the article for credit, follow the prompts associated with the online version of this article.

Editorial Comment on “Utilizing Fully Automated Abdominal CT–Based Biomarkers for Opportunistic Screening for Metabolic Syndrome in Adults Without Symptoms”

This single-center study applied fully automated algorithms to quantify muscle, visceral and subcutaneous fat, liver steatosis, and abdominal aortic calcification. The population consisted of 7785 adults without symptoms who underwent low-dose unenhanced abdominal CT as part of routine virtual colonoscopy. With the use of established clinical criteria, metabolic syndrome was diagnosed in 9.5% of patients. These subjects had a higher incidence of subsequent cardiovascular events. For all four tissues evaluated, the subgroup with metabolic syndrome had significant differences in CT attenuation or the area of attenuation compared with the re-

maining subjects without metabolic syndrome. These valuable study results imply that CT-based algorithms may help alert subjects who have not yet developed symptoms but who have undergone scanning for a variety of reasons about their risk for metabolic syndrome. At the same time, CT measures could be used to improve our knowledge of the multifaceted connections between body composition and metabolic and cardiovascular disease.

As radiologists, we are increasingly asked to provide quantitative data from our imaging modalities. However, to do this, we need quick and user-friendly tools. In busy departments, these tools must